In addition to preforming mathematical calculations very rapidly, DSPs must also have a *predictable* execution time. Suppose you launch your desktop computer on some task, say, converting a word-processing document from one form to another. It doesn't matter if the processing takes ten *milliseconds* or ten *seconds*; you simply wait for the action to be completed before you give the computer its next assignment.

In comparison, most DSPs are used in applications where the processing is *continuous,* not having a defined start or end. For instance, consider an engineer designing a DSP system for an audio signal, such as a hearing aid. If the digital signal is being received at 20,000 samples per second, the DSP must be able to maintain a sustained throughput of 20,000 samples per second. However, there are important reasons not to make it any faster than necessary. As the speed increases, so does the *cost*, the *power consumption*, the *design difficulty*, and so on. This makes an accurate knowledge of the execution time critical for selecting the proper device, as well as the algorithms that can be applied.

Circular Buffering

Digital Signal Processors are designed to quickly carry out FIR filters and similar techniques. To understand the *hardware*, we must first understand the *algorithms*. In this section we will make a detailed list of the steps needed to implement an FIR filter. In the next section we will see how DSPs are designed to perform these steps as efficiently as possible.

To start, we need to distinguish between **off-line processing** and **real-time processing**. In off-line processing, the *entire* input signal resides in the computer at the same time. For example, a geophysicist might use a seismometer to record the ground movement during an earthquake. After the shaking is over, the information may be read into a computer and analyzed in some way. Another example of off-line processing is medical imaging, such as computed tomography and MRI. The data set is acquired while the patient is inside the machine, but the image reconstruction may be delayed until a later time. The key point is that *all* of the information is simultaneously available to the processing program. This is common in scientific research and engineering, but not in consumer products. Off-line processing is the realm of personal computers and mainframes.

In real-time processing, the output signal is produced at the same time that the input signal is being acquired. For example, this is needed in telephone communication, hearing aids, and radar. These applications must have the information immediately available, although it can be delayed by a short amount. For instance, a 10 millisecond delay in a telephone call cannot be detected by the speaker or listener. Likewise, it makes no difference if a radar signal is delayed by a few seconds before being displayed to the operator. Real-time applications input a sample, perform the algorithm, and output a sample, over-and-over. Alternatively, they may input a group